

PAPER_961: Compressed Gravity Triadic Mode

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Abstract

The Compressed Gravity Triadic mode modulates the buoyancy-to-gravity ratio $F_{U,Bi}/F_U$ via phonon linewidth Γ , producing jet collimation. Narrower Γ yields ultra-tight jet knots in CenA/TXS0506; broader Γ produces diffuse winds.

1. Compressed Gravity

$$F_{\text{compressed}}(\Gamma) = \frac{F_{U,Bi}}{F_U} \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26} \cdot A_{\text{jet}}$$

2. Regime Map

Γ (THz)	Regime
< 0.05	Ultra-tight collimation
0.05 - 0.15	Collimated jets
> 0.15	Diffuse wind

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)